Call for Papers

Workshop on Signal Processing for Big Data (SPBD) in Wireless Networks

IEEE Global Telecommunication Conference 2016
Washington DC, Dec. 4 – 8, 2016

Scope and Motivation

With the growing popularity of mobile Internet and internet of things (IoT), connected devices such as cell phones and smart sensors have invaded every aspect of our life and changed the way how we interact and sense the outside world. There is an explosive growth in the amount of data generated by these connected devices. The volume and variety of the data are increasing at an unprecedented pace. Large amount of data impose formidable challenges on the design, operation, and analysis of wireless networks, such as large bandwidth, large storage space, and high energy consumption, etc. Therefore, new signal processing and communication techniques are required to meet the demands for the collections, processing, and transmission of the huge amount of data in wireless networks.

Currently there is a big gap between the resources required to support big data in wireless networks, and the services that can be supplied by current wireless network structures. The collection, transmission, and processing of the huge amount of data demand high energy consumption, large bandwidth, fast processors, and large storage spaces. On the other hand, due to the wireless and mobile nature, most wireless systems need to operate under stringent constraints of various resources, such as small battery, small storage space, limited bandwidth, strong interferences, and limited processor power, etc. Such a big gap between demands and supplies necessitates the development of new theories and practices that can cope with the challenges and opportunities due to big data in wireless networks.

Main Topics of Interests

The objective of this workshop is to bring together academic and industrial researchers to identify and discuss technical challenges and recent results that are related to signal processing for big data in wireless networks. The topics to be covered in this workshop are listed as follows.

- Low-dimensional and sparse signal representations in wireless systems
- Signal processing based on graphic models
- Advanced interference coordination techniques with cloud computing
- Distributed Optimization Algorithms over Wireless Networks
- Resource management (energy, storage, bandwidth, complexity, etc.)
- High-dimensional statistical learning for wireless signals

- Scalable and decentralized optimizations in ad hoc networks
- Randomized algorithms
- Bad data detection/Intrusion detection in sensor networks
- Security and Privacy in Big Data
- Fundamental limits: convergence, consistency, robustness, and performance guarantees

How to Submit a Paper:

Workshop paper can be submitted on EDAS.

All final submissions should be written in English with a maximum paper length of six (6) printed pages (10-point font) including figures without incurring additional page charges (maximum 1 additional page with overlength page charge if accepted).

Important Dates:

- Submission deadline: July 1, 2016, 11:59 pm (NYT)
- Notification deadline: Sept. 1, 2016, 11:59 pm (NYT)
- Camera ready deadline: Oct. 1, 2016, 11:59 pm (NYT)
- Workshop date: Dec. 8, 2018

Organizing Co-chairs:

Jingxian Wu, University of Arkansas
Huaiyu Dai, North Carolina State University
Zhu Han, University of Huston
Lifeng Lai, Worcester Polytechnic Institute